THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 30/09/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High fuel consumption, steep operational expenditures, and critical safety hazards associated with conducting flight training in live jet aircraft.

2. AI RATIONALE & ANALYSIS

"High-fidelity simulators accurately replicate complex jet physics and emergency scenarios on the ground without burning costly aviation fuel or risking aircrews. THEXRA's deployment of MECHALABO provides the turnkey integration of industrial-grade physical simulation and precision telemetry needed to satisfy rigorous aviation training standards."

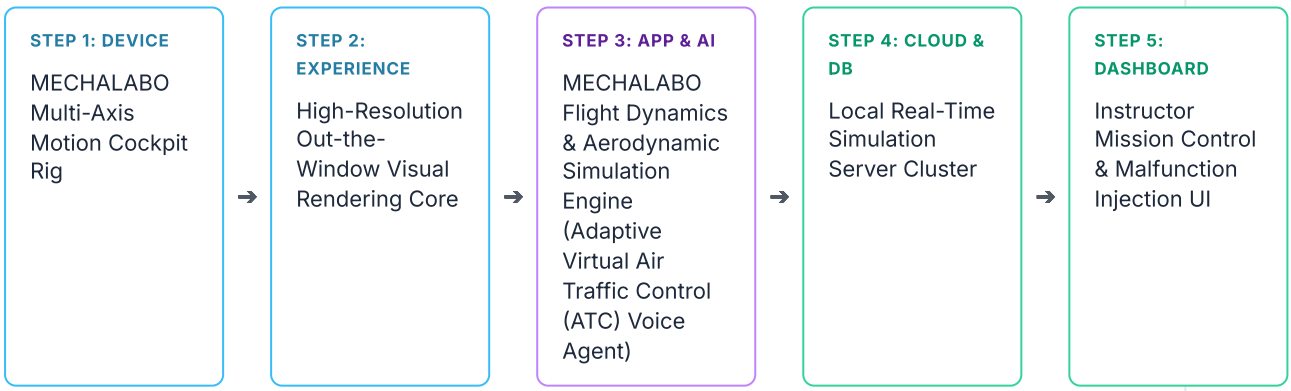
3. PROPOSED SOLUTION CONCEPT

Full-motion high-fidelity jet cockpit simulator platform integrated with real-time aerodynamics physics engines and immersive panoramic projection for hazard and flight procedure training.

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

Simulator — MECHALABO Heavy-Industry Training Simulators

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: A high-fidelity hardware-in-the-loop jet cockpit simulator featuring multi-axis motion cues, accurate flight dynamics modeling, and an instructor operating station. It enables realistic emergency procedure training and instrument navigation drills in a completely safe, fuel-free ground setting.

2. User Journey: Trainee enters the replica cockpit, straps in, and receives briefing parameters from the instructor station. The pilot executes taxi, takeoff, flight maneuvers, and emergency handling routines against real-time physics and environmental hazards. Upon mission completion, trainee and instructor review synchronized flight telemetry and cockpit video logs for debriefing.

3. Hardware: MECHALABO Multi-Axis Motion Cockpit Rig, Collimated Visual Display System / High-FOV Projector Array, Replicated Jet HOTAS Avionics and Tactile Controls, Instructor Operating Station (IOS) Dual-Monitor Console

> **4. Software:** MECHALABO Flight Dynamics & Aerodynamic Simulation Engine, Dynamic Weather and Environmental Hazard Engine, Instructor Control and Fault-Injection Software, Flight Telemetry Logging and Playback Suite
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> **5. AI Components:** Adaptive Virtual Air Traffic Control (ATC) Voice Agent, Automated Pilot Performance Telemetry Scoring Model
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> **6. XR Components:** High-Resolution Out-the-Window Visual Rendering Core, Synthetic Vision Avionics HUD Overlay
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> **7. Backend:** Local Real-Time Simulation Server Cluster, PostgreSQL Telemetry and Flight Log Database, DIS/HLA Military/Aviation Simulation Networking Protocol
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> **8. Dashboard:** Instructor Mission Control & Malfunction Injection UI, Pilot Competency and After-Action Review (AAR) Analytics Portal
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> **9. Data Flow:** Cockpit Controls → Real-Time Simulation Server → Motion Rig & Visual Engines → IOS Dashboard → Analytics DB
>
> **Alternative Option:** Combination solution — MECHALABO

5. EXPECTED BUSINESS OUTCOMES

Up to 80% reduction in jet fuel consumption during procedural training, 100% elimination of live flight hazard risks, and granular post-flight telemetry analytics.

6. IMPLEMENTATION NEXT STEPS

Conduct a technical requirements review with flight operations to benchmark avionics interfaces, cockpit telemetry points, and facility space constraints.